

Started on	Wednesday, 13 November 2024, 1:00 PM
State	Finished
Completed on	Wednesday, 13 November 2024, 1:26 PM
Time taken	25 mins 11 secs
Marks	2.50/3.00
Grade	2.50 out of 3.00 (83.34%)

Information

Information

This page contains all the problems for this test. The very last problem asks you to contact the person in charge of the exam and tell him or her the 4-digit key given in the problem text. In return you will be given a 5-digit signing code which you must give as the answer to the problem.

This problem does not count towards the final score, but **tests missing this code will not count towards the final grade.**

The following rules apply:

- Total time allowed: 30 minutes. The test will automatically close if time runs out.
- UiA's usual rules in regards to cheating on exams apply.

Question 1

Partially correct

Mark 0.50 out of 1.00

The following sequence

$$p_n = -\frac{(-1)^n}{4 \cdot 3^n}$$

is a special solution to the recurrence relation

$$5 \cdot a_n - \frac{(-1)^n}{3^n} + 3 \cdot a_{n-1} = 0.$$

a) Find another solution to the recurrence relation, not equal to p_n :

$$a_n = -((-1)^n/(4 \cdot 3^n)) + (3/5)^n$$

Your last answer was interpreted as follows:

$$-\frac{(-1)^n}{4 \cdot 3^n} + \left(\frac{3}{5}\right)^n$$

The variables found in your answer were: $[n]$

Syntax guide: Your answer should be an expression which depends on the variable n , and *no other constants*. E.g. if you think the answer is

$$-7 \cdot \frac{2^n + 1}{3n},$$

you should type **$-7 \cdot (2^n + 1) / (3 \cdot n)$** .

b) Given the initial condition $a_0 = \frac{7}{4}$. Compute a_n for $n = 1$:

$$a_1 = -(67/60)$$

Your last answer was interpreted as follows:

$$-\frac{67}{60}$$

Question 2

Correct

Mark 1.00 out of 1.00

Let $(D_3, \cdot)$ be the dihedral group of symmetries of the regular triangle. We label the vertices of the triangle 1, 2, 3 in a counter-clockwise manner. Then r_1 is the rotational symmetry by 120 degrees which sends vertex 1 to vertex 2, and r_2 is rotation by 240 degrees. The element t_1 is the reflection symmetry which fixes vertex 1, and likewise for t_2 and t_3 .

The multiplication table for this group looks like

id	r_1	r_2	t_1	t_2	t_3
r_1	r_2	.	.	t_3	.
r_2	id	r_1	t_3	t_1	t_2
t_1	t_3	.	.	r_2	.
t_2	t_1	t_3	r_1	id	r_2
t_3	t_2	.	.	r_1	.

However, we have left out 9 entries in the places marked by dots.

Find the entire multiplication table for D_3 .

$t_1 \cdot t_1 =$	id
$r_1 \cdot t_1 =$	t_2
$t_3 \cdot t_1 =$	r_2
$t_1 \cdot r_2 =$	t_2
$r_1 \cdot r_2 =$	id
$t_3 \cdot r_2 =$	t_1
$t_1 \cdot t_3 =$	r_1
$r_1 \cdot t_3 =$	t_1
$t_3 \cdot t_3 =$	id

Your last answer was interpreted as follows:

id

Your last answer was interpreted as follows:

t_2

Your last answer was interpreted as follows:

r_2

Your last answer was interpreted as follows:

t_2

Your last answer was interpreted as follows:

id

Your last answer was interpreted as follows:

t_1

Your last answer was interpreted as follows:

r1

Your last answer was interpreted as follows:

t1

Your last answer was interpreted as follows:

id

Question **3**

Correct

Mark 1.00 out of 1.00

In this problem you are tasked with computing products and inverses in the group $(P_{49}, \bullet_{49})$ of integers and multiplication modulo 49.

$$36 \bullet_{49} 27 = \boxed{41}$$

Your last answer was interpreted as follows:

41

$$27 \bullet_{49} 32 = \boxed{31}$$

Your last answer was interpreted as follows:

31

$$32 \bullet_{49} 31 = \boxed{12}$$

Your last answer was interpreted as follows:

12

$$36^{-1} = \boxed{15}$$

Your last answer was interpreted as follows:

15

Question **4**

Correct

Mark 0.00 out of 0.00

Signing code

Before closing the test you must answer this problem with a signing code given to you by the person in charge of the test.

Tests missing this signing code will be ignored and will not count towards the final score.

Key: 291

Signing code:

Your last answer was interpreted as follows:

39625

[◀ Test 2 \(topics 4-6: Languages, Machines, Regular expressions, Correct programs, Relations\)](#)

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